GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Engineering Mathematics — Master Formula Sheet & Revision Digest

Essential Mathematical Formulas, Matrix Theorems, Calculus Identities & Distributions

GATE Section: Section 1 • **Marks:** 13 Marks • **Format:** Revision Cheat Sheet

KEY MATHEMATICAL IDENTITIES FOR GATE ECE
- **Eigenvalues:** Sum of eigenvalues = Trace(A), Product of eigenvalues = det(A). If A is symmetric real, all eigenvalues are real.
- **Cauchy-Riemann:** u_x = v_y and u_y = -v_x in Cartesian; u_r = (1/r)*v_theta and v_r = -(1/r)*u_theta in Polar.
- **Residue at pole z0 (order m):** Res(z0) = (1/(m-1)!) * lim_{z->z0} d^(m-1)/dz^(m-1) [(z - z0)^m * f(z)].

SECTION: LINEAR ALGEBRA & MATRIX OPERATIONS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Trace & Determinant	**Tr(A) = sum(lambda_i)** **det(A) = prod(lambda_i)**	lambda_i = eigenvalues of A	Eigenvalues of triangular or diagonal matrix are its diagonal elements.
Cayley-Hamilton Theorem	**P(A) = 0** **A^n + c_{n-1}A^{n-1} + ... + c_0 I = 0**	P(lambda) = characteristic polynomial	Used to compute higher matrix powers A^k and matrix inverse A^-1 without adjoint.
Orthogonal & Unitary	**A^T * A = I (Orthogonal)** **A^H * A = I (Unitary)**	|det(A)| = 1 Eigenvalues have magnitude |lambda| = 1	Length of vector is preserved under orthogonal/unitary transformation.
Rank-Nullity Theorem	**Rank(A) + Nullity(A) = n**	n = number of columns of matrix A	System Ax = 0 has non-trivial solutions iff Rank(A) < n.
Consistency of Ax = b	**Rank(A) = Rank([A|b]) = r**	If r = n -> Unique solution If r < n -> Infinitely many solutions	If Rank(A) < Rank([A|b]) -> Inconsistent (No solution).

SECTION: CALCULUS, MEAN VALUES & VECTOR THEOREMS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Rolle's & LMVT	**f'(c) = [f(b) - f(a)] / (b - a)** **Rolle: f'(c) = 0 if f(a) = f(b)**	c in open interval (a, b)	Function must be continuous on [a, b] and differentiable on (a, b).
Taylor Series	**f(x) = sum_{n=0}^inf [f^(n)(a)/n!] * (x - a)^n**	a = expansion center point	Maclaurin series is Taylor series centered at a = 0.
Euler's Theorem	**x*(df/dx) + y*(df/dy) = n * f(x, y)**	f(x, y) homogeneous of degree n	Used for quick derivative evaluation without direct partial differentiation.
Green's Theorem	**oint (P dx + Q dy) = iint (dQ/dx - dP/dy) dx dy**	Closed counter-clockwise curve C	Converts difficult line integral into simple planar double integral.
Gauss Divergence	**iint_{S} F • dS = iiint_{V} (div F) dV**	div F = dFx/dx + dFy/dy + dFz/dz	Valid for closed surface S bounding volume V. Flux = enclosed divergence.

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Stokes' Theorem	**oint_{C} F • dr = iint_{S} (curl F) • dS**	curl F = del x F	Line integral around boundary C equals surface integral of curl F.

SECTION: DIFFERENTIAL EQUATIONS & COMPLEX ANALYSIS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
1st Order Linear ODE	**dy/dx + P(x)*y = Q(x)** **I.F. = exp(integral P dx)**	Solution: y*(I.F.) = integral (Q * I.F. dx) + C	Non-linear Bernoulli: dy/dx + P*y = Q*y^n transformed using v = y^(1-n).
2nd Order ODE Roots	**m1, m2 distinct: y = C1*e^(m1*x) + C2*e^(m2*x)** **m1=m2: y = (C1 + C2*x)*e^(m1*x)** **alpha +- j*beta:** **e^(alpha*x)(C1*cos(beta*x) + C2*sin(beta*x))**	m = roots of auxiliary equation m^2 + a*m + b = 0	Overdamped, critically damped, and underdamped transient analogs in circuits.
Cauchy Integral Formula	**oint [f(z) / (z - z0)^(n+1)] dz = [2*pi*j / n!] * f^(n)(z0)**	z0 inside contour C, f(z) analytic on and inside C	Zero if z0 lies OUTSIDE the contour C (by Cauchy-Goursat theorem).
Residue Theorem	**oint f(z) dz = 2 * pi * j * sum(Res(z_k))**	z_k = poles enclosed inside contour C	Used to evaluate difficult real improper integrals from -inf to +inf.

SECTION: PROBABILITY & DISCRETE / CONTINUOUS DISTRIBUTIONS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Bayes' Theorem	**P(A_i | B) = [P(B | A_i)*P(A_i)] / sum[P(B | A_k)*P(A_k)]**	A_i form mutually exclusive & exhaustive partition	Crucial for digital communications symbol detection & MAP decoding.
Binomial Distribution	**P(X = k) = nCk * p^k * (1 - p)^(n - k)** **Mean = n*p, Var = n*p*(1 - p)**	n trials, p = success probability	Approaches Poisson when n -> inf, p -> 0 with lambda = n*p constant.
Poisson Distribution	**P(X = k) = (lambda^k * e^(-lambda)) / k!** **Mean = lambda, Var = lambda**	lambda = average arrival rate	Variance equals Mean! Frequently tested in packet arrival questions.
Normal (Gaussian)	**f(x) = (1 / (sigma*sqrt(2*pi))) * exp(-(x - mu)^2 / (2*sigma^2))** **Q(k) = P(X > mu + k*sigma)**	mu = mean, sigma^2 = variance	68.3% within mu +- sigma, 95.4% within mu +- 2*sigma, 99.7% within mu +- 3*sigma.
Exponential Distribution	**f(x) = lambda * exp(-lambda * x), x >= 0** **Mean = 1/lambda, Var = 1/lambda^2**	Memoryless property: P(X > s + t | X > s) = P(X > t)	Only continuous distribution possessing the memoryless property.